

Math 526, F06
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	10	
2	15	
3	20	
4	20	
5	15	
6	20	
Total	100	

Good Luck !

Problem 1. (10 points) Let $A = \begin{bmatrix} 0 & -2 & 2 \\ -2 & 1 & -3 \\ 7 & 4 & -1 \\ -1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 4 & -4 \end{bmatrix}$.

a) Which one of the products AB^t and A^tB is possible? Explain.

Solution: A is 4×3 matrix $\Rightarrow A^t$ is 3×4 matrix

B is 2×3 matrix $\Rightarrow B^t$ is 3×2 matrix

So AB^t is possible and results in a 4×2 matrix

b) Compute the product from a) that is possible.

Solution: $B^t = \begin{bmatrix} 1 & -1 \\ -2 & 4 \\ 3 & -4 \end{bmatrix}$

$$\Rightarrow AB^t = \begin{bmatrix} 10 & -16 \\ -13 & 18 \\ -4 & 13 \\ 4 & -3 \end{bmatrix}$$

Problem 2. (15 points)

a) Find two 2×2 matrices A and B such that $AB = 0_{2 \times 2}$, but $A \neq 0_{2 \times 2}$ and $B \neq 0_{2 \times 2}$.

Solution: There are infinitely many such two 2×2 matrices.

For example:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

b) Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 4 & 12 \\ -2 & -4 & -8 \end{bmatrix}$. Can we find a 3×3 matrix B such that $AB = I$? Explain.

Solution: We first find L-U factorization of A .

$$\begin{matrix} L & A \\ \left[\begin{array}{c} \\ \\ \end{array} \right] & \begin{bmatrix} 1 & 2 & 4 \\ 3 & 4 & 12 \\ -2 & -4 & -8 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} & U \\ \left[\begin{array}{c} 3 \\ -2 \end{array} \right] & \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$\Rightarrow A = LU = \begin{bmatrix} 1 & & \\ 3 & 1 & \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

So A is singular due to the fact that U is singular. Therefore, A doesn't have an inverse B such that $AB = I$.

Problem 3. (20 points)

a) Consider the 2×3 matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}, \quad C = \begin{bmatrix} 2a_{11} - 2a_{21} & 2a_{12} - 2a_{22} & 2a_{13} - 2a_{23} \\ 7a_{11} & 7a_{12} & 7a_{13} \\ -3a_{11} + a_{21} & -3a_{12} + a_{22} & -3a_{13} + a_{23} \end{bmatrix}.$$

Find a 3×2 matrix B so that A multiplied by B equals C . From which side should you multiply A by B ?

Solution: $B = \begin{bmatrix} 2 & -2 \\ 7 & 0 \\ -3 & 1 \end{bmatrix}$

B should be multiplied on the left
such that

$$BA = C$$

b) How many flops does it take to compute the product of a tridiagonal $n \times n$ matrix and a column n -vector if multiplication by zero is not executed? Explain.

Solution: $5(n-2) + 6 = 5n - 4$

Problem 4. (20 points) Let $A = \begin{bmatrix} 1 & -1 & 3 \\ -2 & 4 & -1 \\ 3 & 3 & 4 \end{bmatrix}$.

a) Find the L - U - P factorization of A using the strategy of partial pivoting.

Solution:

$$\begin{matrix} L & U & P \\ \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} & \begin{bmatrix} 1 & -1 & 3 \\ -2 & 4 & -1 \\ 3 & 3 & 4 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} 3 & 3 & 4 \\ -2 & 4 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ -\frac{2}{3} & & \\ \frac{1}{3} & & \end{bmatrix} \begin{bmatrix} 3 & 3 & 4 \\ 0 & 6 & \frac{5}{3} \\ 0 & -2 & \frac{5}{3} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} & & \\ -\frac{2}{3} & & \\ \frac{1}{3} & -\frac{1}{3} & \end{bmatrix} \begin{bmatrix} 3 & 3 & 4 \\ 0 & 6 & \frac{5}{3} \\ 0 & 0 & \frac{20}{9} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow L = \begin{bmatrix} 1 & & & \\ -\frac{2}{3} & 1 & & \\ \frac{1}{3} & -\frac{1}{3} & 1 & \end{bmatrix} \quad U = \begin{bmatrix} 3 & 3 & 4 \\ 0 & 6 & \frac{5}{3} \\ 0 & 0 & \frac{20}{9} \end{bmatrix} \quad P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

b) Find the L - D - U - P factorization ($LDU = PA$) of A using the result from a).

$$\begin{matrix} L & D & U & P & A \\ \begin{bmatrix} 1 & & & \\ -\frac{2}{3} & 1 & & \\ \frac{1}{3} & -\frac{1}{3} & 1 & \end{bmatrix} & \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & \frac{20}{9} \end{bmatrix} & \begin{bmatrix} 1 & 1 & \frac{4}{3} \\ 0 & 1 & \frac{5}{18} \\ 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ -2 & 4 & -1 \\ 3 & 3 & 4 \end{bmatrix} \end{matrix}$$

Problem 5. (15 points) Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix}$.

a) Explain why A is invertible.

Solution: A is invertible since all its diagonal components are nonzero, i.e., A is nonsingular.

b) Find the inverse matrix of A .

Solution: The first column of A^{-1} is found by

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

The second column of A^{-1} is found by

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}$$

The third column of A^{-1} is found by

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} \\ -\frac{3}{8} \\ \frac{1}{4} \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 1 & -1 & \frac{1}{4} \\ 0 & \frac{1}{2} & -\frac{3}{8} \\ 0 & 0 & \frac{1}{4} \end{bmatrix}$$

Problem 6. (20 points) Let $A = \begin{bmatrix} 4 & -2 & & \\ -2 & 4 & 1 & \\ & -1 & 5 & 2 \\ & & 2 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} -4 \\ 0 \\ -6 \\ 0 \end{bmatrix}$.

a) Can the tridiagonal solver be applied to the system $Ax = b$? Explain.

Solution: Yes.

i) A is column diagonally dominant.

ii) All super-diagonal components of A are nonzeros.

iii) $|a_{11}| = 4 > 2 = |a_{21}|$

b) Find the solution of the above system in a) using the tridiagonal solver.

Solution:

$$\begin{bmatrix} 4 & -2 & & \\ -2 & 4 & 1 & \\ & -1 & 5 & 2 \\ & & 2 & 2 \end{bmatrix} \Rightarrow L = \begin{bmatrix} 1 & & & \\ -\frac{1}{2} & 1 & & \\ & -\frac{1}{3} & 1 & \\ & & \frac{3}{8} & 1 \end{bmatrix} \quad U = \begin{bmatrix} 4 & -2 & & \\ & 3 & \frac{1}{3} & \\ & & \frac{16}{3} & 2 \\ & & & \frac{5}{4} \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 4 & -2 & & \\ -\frac{1}{2} & 3 & 1 & \\ & -1 & 5 & 2 \\ & & 2 & 2 \end{bmatrix} \Rightarrow Ly = b = \begin{bmatrix} -4 \\ 0 \\ -6 \\ 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 4 & -2 & & \\ \textcircled{-\frac{1}{2}} & 3 & \frac{1}{3} & \\ & \textcircled{\frac{1}{3}} & \frac{16}{3} & 2 \\ & & 2 & 2 \end{bmatrix} \Rightarrow y = \begin{bmatrix} -4 \\ -2 \\ -\frac{20}{3} \\ \frac{5}{2} \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 4 & -2 & & \\ \textcircled{-\frac{1}{2}} & 3 & \frac{1}{3} & \\ & \textcircled{-\frac{1}{3}} & \frac{16}{3} & 2 \\ & & \textcircled{\frac{3}{8}} & \frac{5}{4} \end{bmatrix} \Rightarrow Ux = y = \begin{bmatrix} -4 \\ -2 \\ -\frac{20}{3} \\ \frac{5}{2} \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 4 & -2 & & \\ \textcircled{-\frac{1}{2}} & 3 & \frac{1}{3} & \\ & \textcircled{-\frac{1}{3}} & \frac{16}{3} & 2 \\ & & \textcircled{\frac{3}{8}} & \frac{5}{4} \end{bmatrix} \Rightarrow x = \begin{bmatrix} -1 \\ 0 \\ -2 \\ 2 \end{bmatrix}$$